Avinash Maskey

Kathmandu, Nepal

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Research Interests

Causal analysis of information systems and digital platforms; understanding how system design, policy and algorithmic interventions shape user behavior and outcomes; empirical analysis of fintech systems, e-commerce and educational technology systems; and data-informed decision making in complex socio-technical environments.

Education

Master of Computer Information System (MCIS) March 2020 – February 2024

Nepal College of Information Technology, Pokhara University, Nepal

CGPA: 3.88 / 4.00 (Batch Topper); University Topper in 3rd Semester (4.00 / 4.00)

Thesis: *Forecasting Stock Exchange Sector Indices Using Granger Causality and Vector Autoregressive Model*

Supervisor: Dr. Suresh Pokharel, PhD

Bachelor of Information Management (BIM) October 2014 – October 2018

Kist College of Management, Tribhuvan University, Nepal

CGPA: 3.61 / 4.00

University Top-5 ranking in 5th and 8th semesters; College semester topper in 5th, 6th and 8th semesters

Publications and Research Experience

Peer-Reviewed Journal Publications

Shrestha, N., Manandhar, N. K., Bhandari, B., **Maskey, A.**, & Giri, P. R. (2026). Undergraduate students' perceptions of generative artificial intelligence (Gen AI) in academic assignments. *Advances in Mobile Learning Educational Research, 6*(1), 1809–1823. doi:10.25082/AMLER.2026.01.014

- Co-authored a Technology Acceptance Model-informed quantitative study of 174 undergraduates using descriptive statistics, independent-samples t-tests and one-way ANOVA.
- Found that 63.8% used generative AI daily and reported positive perceptions of its academic use ($M = 3.82$, $SD = 0.44$); adoption was shaped by ease of use, instructor integration, mobile access and institutional support.

Maskey, A. (2026). Resilience of Nepal's stock market to systemic shocks: A case study of COVID-19. *SEBON Journal, 12*(1), 59–92.

- Applied Double Machine Learning with Ridge regression and five-fold cross-fitting to 1,371 NEPSE trading days from 2018–2023, controlling for macroeconomic, pandemic-policy and market variables.
- Estimated approximately 21 basis points higher average daily returns during COVID-19 than the macroeconomic counterfactual and tested robustness using alternative specifications, GARCH, event studies, sensitivity analysis and 1,000 placebo permutations.

Conference Presentation

Stock Market Awareness and Participation of University Students: A Case of Kathmandu University

1st Joint National Conference on Capital Market in the AI Era, Nepal 2026

- Presented a university-wide survey of 153 students examining stock market awareness, participation and trading behavior using logistic and linear regression.
- Found that highly aware students were six times more likely to participate and five times more likely to trade; awareness explained 55.2% of participation intensity, with classification accuracies of 80.4% and 86.3%.

Master's Thesis

Forecasting Stock Exchange Sector Indices Using Granger Causality and Vector Autoregressive Models

Pokhara University 2024

- Developed an adaptive VAR model with an empirically selected window that outperformed static and fixed rolling-window models, particularly for development banks, finance companies and hydropower.
- Used Granger causality and impulse response analysis to trace cross-sector shock transmission and identify banking as a major source of market volatility.

Academic Teaching and Assessment Experience

Kathmandu University, School of Education (STEAM Department) Kathmandu, Nepal
Lecturer (Full-Time) July 2025 – Present

- Teach undergraduate and graduate courses including Database Systems, Web Applications, System Analysis and Design, Software Engineering, E-Commerce, Management Information Systems and AI in STEAM Education.
- Supervise student projects and internships and evaluate performance through examinations, assignments and project assessments.

Curriculum Designer October 2024 – Present

- Design Kathmandu University curricula and assessment frameworks for BTech Ed IT subjects including Professional IT Project-II, E-Governance, Database System and IT Internship – Work Based Learning.

Lecturer (affiliated colleges under Tribhuvan and Kathmandu University) Kathmandu, Nepal
Lecturer November 2019 – July 2025

- Taught courses including Web Technology I and II, Software Engineering, Database Management Systems, System Analysis and Design, E-Commerce, E-Governance and Ethics in IT.
- Contributed to board examination question setting, internal assessment moderation and academic evaluation processes.

Industry Experience

Prajanya Soft Pvt. Ltd. Kathmandu, Nepal
Director of Product Management January 2022 – June 2025

- Lead the end-to-end development of web and mobile systems across fintech, healthcare and consumer service domains.
- Oversee the design and delivery of an e-commerce platform with inventory management, order workflows and responsive user interfaces, balancing usability with operational reliability.
- Directed the development of a hospital management ERP system covering patient records, diagnostics, billing and inventory, with emphasis on data integrity and role-based access.
- Managed an international client project for an Australia-based trade services platform, coordinating requirements, system architecture decisions and delivery across distributed teams.
- Work closely with engineers, designers and QA professionals to translate business processes into maintainable backend systems and user-facing workflows while making prioritization and release decisions.

Kriti Capital & Investments Limited (now Machhapuchchhre Capital) Kathmandu, Nepal
Financial Analyst – Trading and Research Department March 2019 – July 2019

Century Commercial Bank Limited (now Prabhu Bank) Kathmandu, Nepal
IT Intern – IT Department April 2018 – August 2018

Technical Skills

- Programming: Python, Java, PHP, JavaScript, R
- Python Libraries: pandas, NumPy, scikit-learn, statsmodels, pmdarima, matplotlib, causalgraphicalmodels
- Databases and Version Control: MySQL, Git, GitHub, GitLab
- Project and Product Tools: Jira, Confluence, Trello, Asana
- Digital Analytics Tools: Google Analytics (GA4), Google Search Console, Google Keyword Planner, Google Ads, Meta Business Suite, YouTube Analytics
- Languages: English, Nepali, Hindi

Awards and Honors

- Batch Topper, Master of Computer Information System, NCIT.
- University Top-5 Rank Holder, BIM program.
- President, IT Club and Sports Club, Kist College of Management.

References

Academic references available upon request.